

Data Vortex 2026 - Round 2 Technical Report

1. Problem Definition

The Social Engine requires a semantic layer capable of interpreting short human-language posts.

This project builds two classifiers:

```
post_text → sentiment_label  
post_text → topic_category
```

The sentiment model predicts Positive, Negative, or Neutral. The topic model predicts the category of the post.

2. Dataset Overview

The dataset contains 9,000 records and four source columns:

```
text_id  
post_text  
sentiment_label  
topic_category
```

The inspection found no missing values, blank text values, duplicate complete rows, or duplicate text IDs.

There are 7,900 unique text values and 987 repeated text values. Repeated texts have no conflicting labels.

The sentiment labels are balanced with 3,000 records per class. The topic labels are strongly imbalanced.

3. Preprocessing Pipeline

The preprocessing pipeline:

1. Decodes HTML entities.
2. Decodes escaped Unicode sequences where possible.
3. Replaces URLs with `URLTOKEN` .
4. Replaces mentions with `USERTOKEN` .
5. Normalizes whitespace.
6. Converts text to lowercase.
7. Creates word-level unigram/bigram and character n-gram TF-IDF features.

The sentiment pipeline combines word and character TF-IDF features. The topic pipeline uses character n-grams with balanced class weights. Both use sublinear term frequency and regularized Logistic Regression.

4. Model Selection

Multiple TF-IDF configurations and Logistic Regression settings were compared:

- Word-level TF-IDF.
- Character-level TF-IDF.
- Combined word and character TF-IDF.

Configurations were selected using validation macro F1-score because macro F1 gives equal importance to every class.

Selected sentiment model: **combined_tfidf_logistic_regression**

Selected topic model: **char_tfidf_logistic_regression**

Final models were saved as:

```
round2/models/sentiment_model.pkl  
round2/models/topic_model.pkl
```

5. Training Methodology

Split	Records
Training	5382
Validation	1805
Testing	1813

A group-aware split used the original text as the group. Therefore identical texts could not cross dataset splits.

Train/validation overlap: 0

Train/test overlap: 0

Validation/test overlap: 0

The selected model was retrained on training plus validation records and evaluated once on the untouched test set.

6. Evaluation Metrics

Sentiment classification

Metric	Score
Accuracy	0.6062
Macro precision	0.6069
Macro recall	0.6065
Macro F1-score	0.6067
Weighted F1-score	0.6063

Per-class results:

Class	Precision	Recall	F1-score	Support
Negative	0.6622	0.6501	0.6561	603
Neutral	0.5456	0.5474	0.5465	612
Positive	0.6129	0.6221	0.6174	598

Topic classification

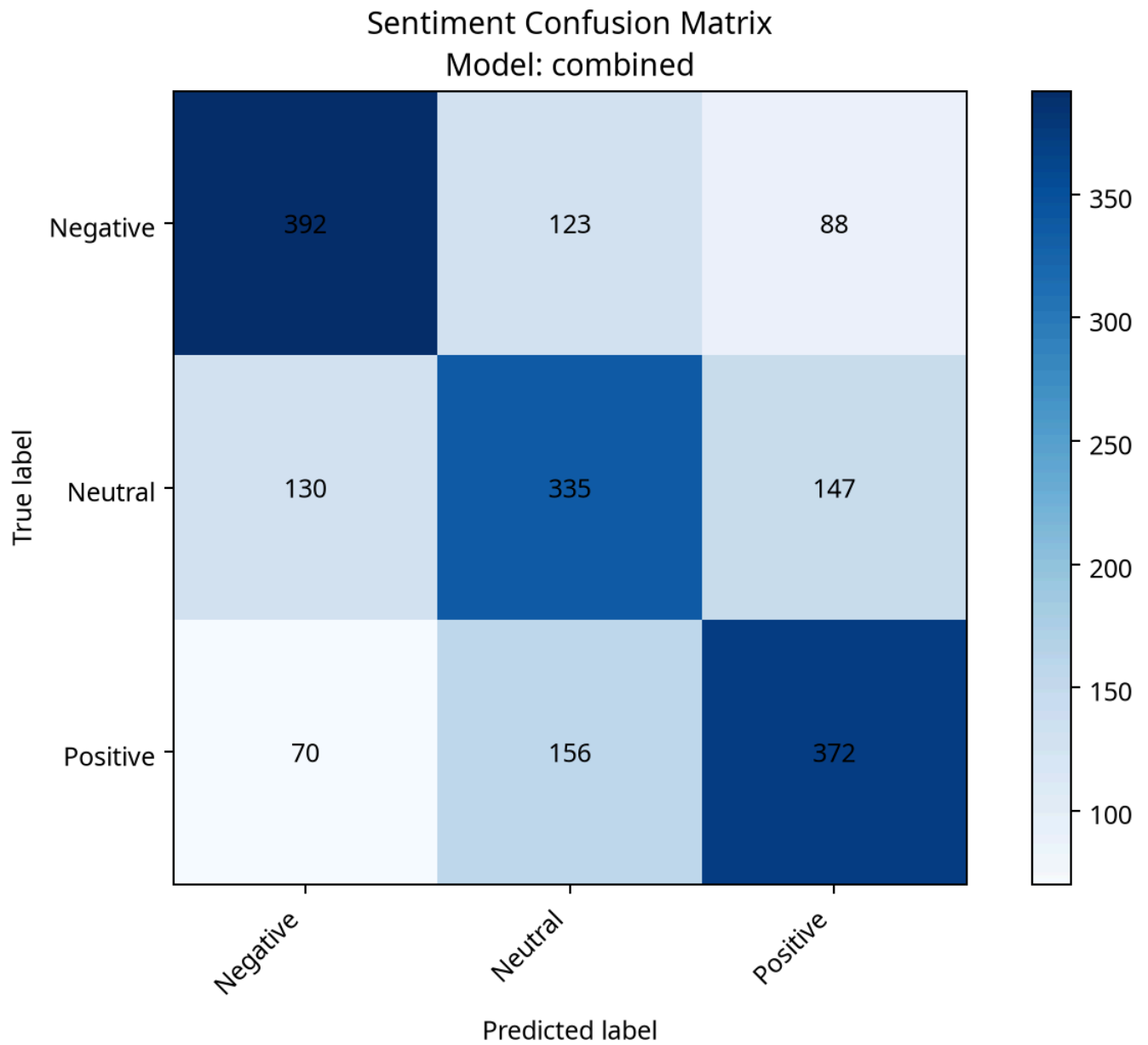
Metric	Score
Accuracy	0.9619
Macro precision	0.9157
Macro recall	0.7607
Macro F1-score	0.8241
Weighted F1-score	0.9597

Per-class results:

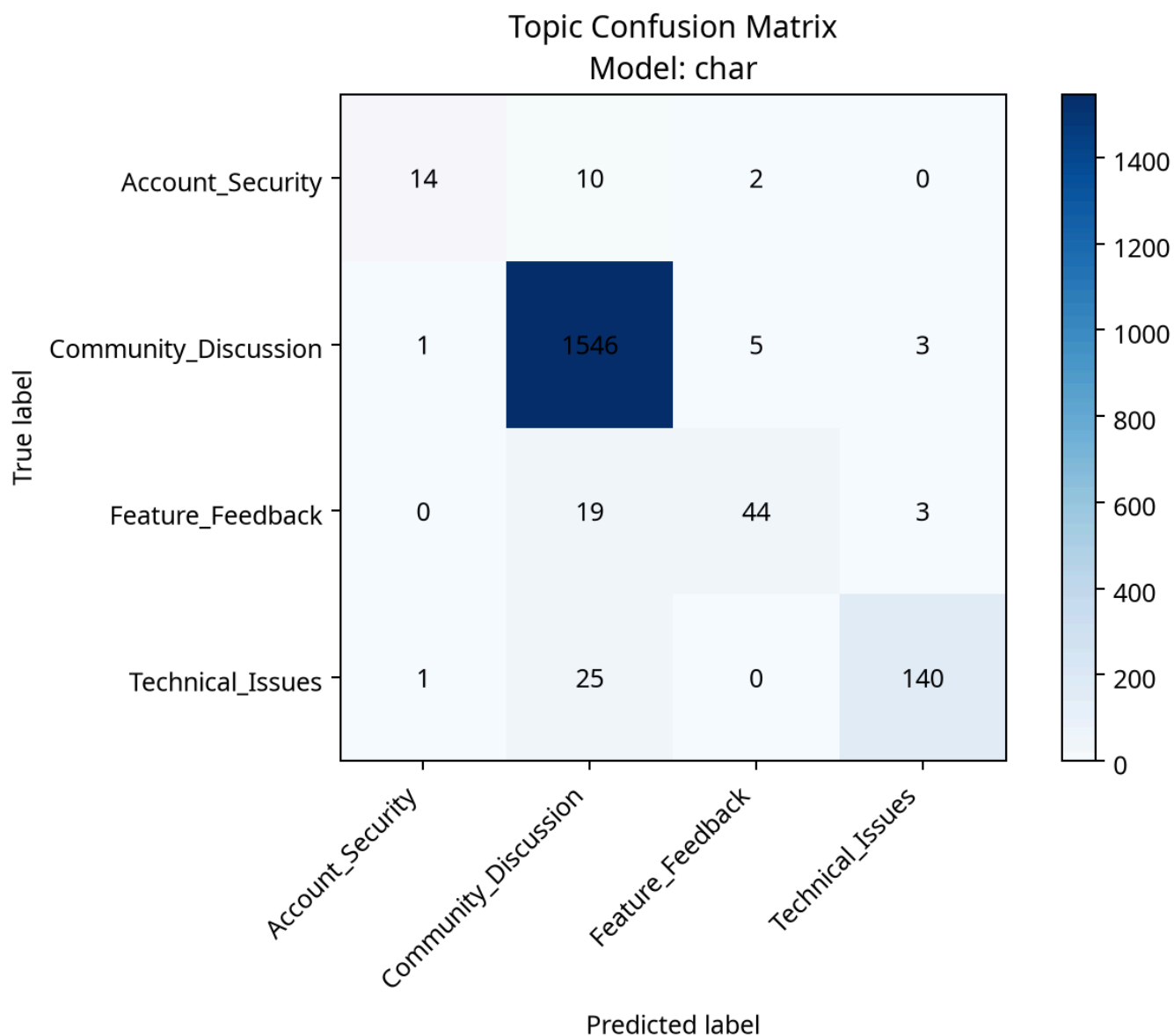
Class	Precision	Recall	F1-score	Support
Account_Security	0.8750	0.5385	0.6667	26
Community_Discussion	0.9663	0.9942	0.9800	1555
Feature_Feedback	0.8627	0.6667	0.7521	66
Technical_Issues	0.9589	0.8434	0.8974	166

7. Confusion Matrices

Sentiment



Topic



The matrices show strong majority-class performance and more difficulty with neutral sentiment and minority topics.

8. Error Analysis

The error files were generated from final test predictions:

```
round2/reports/sentiment_errors.csv  
round2/reports/topic_errors.csv
```

Sentiment error examples

Text ID	Actual label	Predicted label	Text
TXT_00003	Positive	Neutral	Would you like to join us at our annual gala at the Sandman Signature Resort on Oct 25? Contact our centre at 6042773100 for tickets!
TXT_00010	Neutral	Positive	“1st game played under Solar Powered floodlights in the world happened in MYSA,Nairobi Kenya. #PhilipsLED”
TXT_00092	Negative	Neutral	I may have just dropped two letter grades of intelligence after listening to that Kanye West rant...
TXT_00108	Negative	Neutral	November 21 in the lonely hour tour.. OKAY SAM SMITH OKAY!!!! :-((((TAPOS SOLD OUT PA OKAY!!!!!! OKAY LANG TALAGA
TXT_00131	Neutral	Positive	@user Milan, in my opinion, have only the 7th best squad, after Inter, Roma, Juventus, Lazio, Napoli and Fiorentina.”

Typical causes include sarcasm, slang, mixed sentiment, short text, and neutral language containing emotional words.

Topic error examples

Text ID	Actual label	Predicted label	Text
TXT_00079	Account_Security	Community_Discussion	bloodymary - hackers nirvana - smell like teen spirit butterfingers - nicotine feeder - 7 days in the sun smashing pumpkins - bullets with..
TXT_00296	Technical_Issues	Community_Discussion	SUN UPDATE: UFC 191, Beyonce with Ronda Rousey, Jimmy Snuka, TUF, New WWE stable, CHIKARA King of Trios
TXT_00643	Feature_Feedback	Community_Discussion	I don't see why justin thinks it's okay to work Tuesday's.. It's fall were supposed to get \$1 burgers at bar Louie and blackout
TXT_00807	Account_Security	Community_Discussion	Will Fleeing Syrians Flood the Food banks or be served at The Ritz ? David Cameron Merkel Duncan Smith Theresa May The Pope Sentimentalists
TXT_00820	Account_Security	Community_Discussion	"may the gods speed him, RIP, Frank Gifford, New York Giants legend and husband to Kathy Lee Gifford, dead at 84

Topic errors are concentrated in minority categories because those categories contain fewer training examples and share vocabulary with other topics.

9. Limitations

The topic dataset is highly imbalanced. The model performs better on Community Discussion than on the smallest classes.

TF-IDF cannot fully understand sarcasm, outside context, world knowledge, or subtle meaning.

Future improvements could include character features, transformer embeddings, cross-validation, hyperparameter tuning, and additional minority-class data.

10. Reproducibility

Run these commands from the project root:

```
python "round2\src\inspect_dataset.py"
python "round2\src\check_text_duplicates.py"
python "round2\src\train_models.py"
python "round2\src\generate_reports.py"
```

11. Conclusion

This project rebuilds the Social Engine semantic layer using text normalization, duplicate-aware splitting, TF-IDF features, model comparison, final test evaluation, confusion matrices, and error analysis.

The resulting system performs sentiment recognition and topic classification with transparent evaluation and documented limitations.